

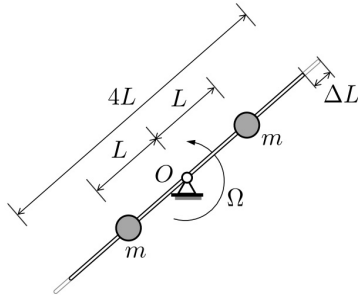
Problems Week 14

~lschen

Problem 1: Rotating Bar-Particle System I

A rod of negligible mass and length $4L$ rotates about its midpoint O with a constant angular velocity Ω . The cross-sectional area of the bar is A , its Young's modulus is E . Two particles of equal masses m are attached to the bar at distances L from the hinge O , as shown below. Neglect gravity.

Given: m, L, A, E, Ω



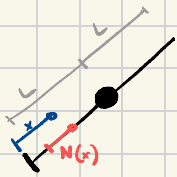
What is the elongation ΔL at the tip on each side of the bar?

Hints:

$$\sigma = E \cdot \varepsilon = \frac{N}{A}$$

for 1D: $\varepsilon = \frac{du}{dx}$

$\Omega = \text{const} \Rightarrow$ no time-dependence!



Divide into two sectors $\Rightarrow$
(just like Mechanics II)
 $0 \leq x < L$
 $L \leq x \leq 2L$

Approach 1: Consider rotating frame M

$\hookrightarrow$ no a''

$\hookrightarrow$ centrifugal force: $F_{\text{centrifugal}} = mL\Omega^2 \hat{e}_x$

$$\Rightarrow N(x) = mL\Omega^2, \text{ if } 0 \leq x < L$$

$$\Rightarrow N(x) = 0, \text{ if } L \leq x \leq 2L$$

Approach 2: inertial perspective & use polar coordinates

$$m a_r = -N(x) \rightarrow a_r = -x_{\text{cm}} \Omega^2 = -L \Omega^2 \Rightarrow \text{same as before}$$

$$\rightarrow \sigma(x) = \frac{N(x)}{A} = E \frac{du(x)}{dx}$$

$$\Rightarrow \frac{N(x)}{EA} = \frac{du}{dx}$$

$$0 \leq x < L: \frac{mL\Omega^2}{EA} = \frac{du}{dx}$$

$$\int dx \Rightarrow u(x) = \frac{mL\Omega^2}{EA} x + c_1$$

$$L \leq x \leq 2L: 0 = \frac{du}{dx}$$

$$\int dx \Rightarrow u(x) = c_2$$

IC: $u(0) = 0$ & $u(x)$ must be continuous at $x = L$

$$\Rightarrow u(x) = \frac{mL\Omega^2}{EA} x, \quad u(x) = \frac{mL^2\Omega^2}{EA}$$

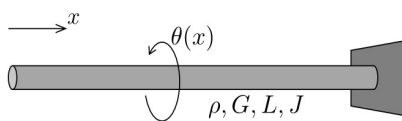
$$0 \leq x < L \quad L \leq x \leq 2L$$

$$\left\{ \begin{array}{l} u(2L) = \frac{m\Omega^2 L^2}{EA} \end{array} \right.$$

Problem 2: Vibration of a Free-Clamped Torsional Shaft

Consider a slender elastic torsional shaft of length L , shear modulus G , mass density ρ , and polar moment of inertia J . The shaft is clamped at $x = L$ and is free to rotate along its axis at $x = 0$, as shown below. Denote by $\theta(x, t)$ the twist angle of the cross-section at position x along the shaft at time t . Neglect gravity.

Given: ρ, G, L, J



What is the general solution $\theta(x, t)$ for free vibrations of the shaft? What is its fundamental frequency ω_0 ?

$$\ddot{\Theta}(x, t) = c_T^2 \Theta_{,xx}(x, t), \quad \text{where } c_T = \sqrt{\frac{G}{\rho}}$$

$$\text{ICs: } \Theta_{,x}(0, t) = 0 \quad \forall t \rightarrow \text{free end} \Rightarrow \text{torque-free}$$

$$\Theta(L, t) = 0 \quad \forall t \rightarrow \text{clamped} \Rightarrow \text{no displacement}$$

$$\hookrightarrow \text{using separable Ansatz: } \Theta(x, t) = \hat{\Theta}(x) \cdot q(t)$$

our ICs are independent on time hence we just look at $\hat{\Theta}(x)$

$$\hat{\Theta}(x) = B_1 \cos\left(\frac{\omega}{c_T} x\right) + B_2 \sin\left(\frac{\omega}{c_T} x\right)$$

$$\hat{\Theta}_{,x}(0) = B_2 \frac{\omega}{c_T} \cos\left(\frac{\omega}{c_T} \cdot 0\right) - B_1 \frac{\omega}{c_T} \sin\left(\frac{\omega}{c_T} \cdot 0\right) \stackrel{!}{=} 0 \Rightarrow B_2 = 0$$

$$\hat{\Theta}(L) = B_1 \cos\left(\frac{\omega}{c_T} \cdot L\right) \stackrel{!}{=} 0 \Rightarrow \cos\left(\frac{\omega}{c_T} \cdot L\right) = 0$$

$$\hookrightarrow \frac{\omega}{c_T} \cdot L = \left(n + \frac{1}{2}\right) \pi, \quad n = \mathbb{N}_0$$

$$\hookrightarrow \hat{\Theta}(x) = B_{1,n} \cdot \cos\left(\left(n + \frac{1}{2}\right) \pi \cdot \frac{x}{L}\right)$$

$$\Rightarrow q(t) = A_{1,n} \cos\left(\left(n + \frac{1}{2}\right) \pi \frac{c_T t}{L}\right) + A_{2,n} \sin\left(\left(n + \frac{1}{2}\right) \pi \frac{c_T t}{L}\right)$$

$$\Rightarrow \Theta(x, t) = \sum_{n=0}^{\infty} B_{1,n} \cos\left(\left(n + \frac{1}{2}\right) \frac{\pi x}{L}\right) \left[A_{1,n} \cos\left(\left(n + \frac{1}{2}\right) \pi \frac{c_T t}{L}\right) + A_{2,n} \sin\left(\left(n + \frac{1}{2}\right) \pi \frac{c_T t}{L}\right) \right]$$

other constants cannot be found with given info!